

Interpretable machine learning and graph attention network based model for predicting PAMPA permeability

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ABSTRACT

Parallel artificial membrane permeability assay (PAMPA) is widely used in the early phases of drug discovery as it is quite robust and offers high throughput. It serves as a platform for assessing the permeability and absorption of pharmaceutical compounds across lipid membranes. This study uses machine learning (Random forest or RF, Explainable boosting machine or EBM and Adaboost) and deep learning (Graph attention network or GAT) to build models to predict PAMPA permeability. A curated dataset of 5447 compounds with PAMPA permeability scores (in a scale 10^{-6} cm/s) was used to train and validate these models. During validation it was observed that, RF and EBM models could predict with an accuracy of 81 % and 80 % respectively, whereas with Adaboost and GAT, the accuracies were limited 76 % and 74 % respectively. Further, an external dataset was used to screen the predictive capability of these models and results showed that RF, EBM and Adaboost had quite similar accuracies with 91 %, 90 % and 89 % respectively. Interestingly, with this external dataset, the GAT-based model also reached a significant accuracy of 86 %. The overall results show that all the models in this study could well predict PAMPA permeability over the benchmark and covering diverse chemical space. All the datasets and codes for developing these models have been deposited on the GitHub platform (https://github.com/hridoy69/pampa_premeability).

1. Introduction

Parallel artificial membrane permeability assay (PAMPA) is a widely recognized in vitro method used to evaluate the permeability of compounds across a synthetic membrane. Originally devised by Kansy et al. [1], PAMPA utilizes a two-compartment system, donor and acceptor compartment, that are separated by a lipid-infused membrane, typically made of lecithin dissolved in an organic solvent. By measuring the transfer rate of compounds from the donor to the acceptor compartment, PAMPA serves as a rapid and cost-effective tool for estimating the permeability potential of drug candidates and other chemicals. Its high-throughput capability and robust design make PAMPA an essential tool in early-stage drug discovery, allowing for the efficient screening of numerous compounds [2,3].

In-silico models have been reported on using PAMPA for predicting membrane permeability based on permeability scores such as $\log_{pampa}$,

$\log P_{eff}$, along with permeation dynamics [4–6]. Machine learning (ML) and Quantitative structure–activity relationship (QSAR) models have also been reported of using PAMPA assays for predicting passive permeation, cell permeation, oral and gastro intestinal absorption of drug molecules [7–12]. In a recent work, Williams et al. [13] have demonstrated using ML and Graph convolution neural networks (GCNN) based models to classify low PAMPA permeability compounds from those with high or moderate PAMPA permeability. They used the PAMPA pH 5 dataset with structural features and relevant descriptors in building the models. Another aspect where PAMPA assay has been widely studied in passive permeability of blood–brain barrier (BBB) of drugs. Four PAMPA models were evaluated by Mensch et al. in two different studies to find the most discerning method for predicting BBB permeability using PAMPA features such as P_{eff} , flux and membrane retention. They also studied how integration of LogBB data with PAMPA information, physicochemical and structural parameters can improve

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Table 1

Summary of the dataset used in model development in this study.

	Training set		Hold-out test set	
Class distribution	Class 1	Class 0	Class 1	Class 0
No. of compounds	2332	2025	583	507
Dataset size	4357		1090	

Table 2

Performance metrics used in this study for evaluating the models.

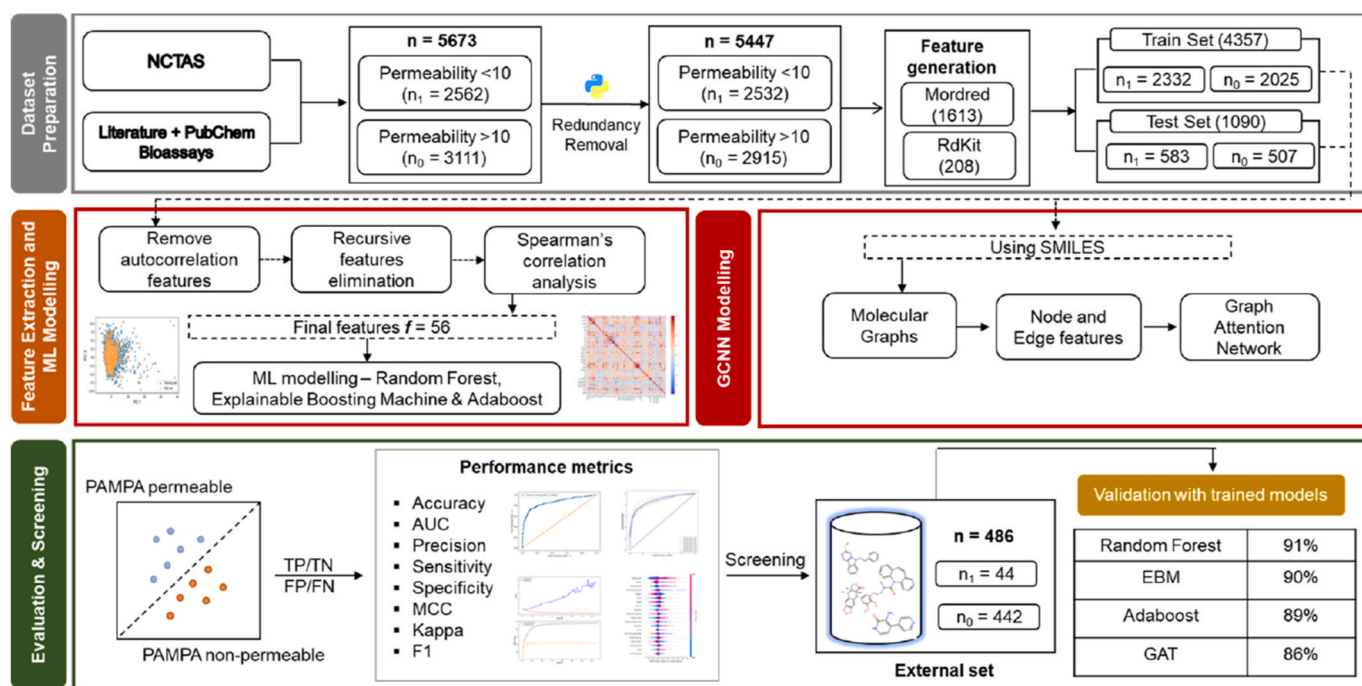
Evaluation metrics	Mathematical formula
Accuracy	$\frac{\text{No. of correct predictions}}{\text{Total no. of predictions}}$
Sensitivity (Recall)	$\frac{TP}{TP + FN}$
Specificity	$\frac{TN}{TN + FP}$
Balanced accuracy	$\frac{1}{2} \left(\frac{TP}{TP + FN} + \frac{TN}{TN + FP} \right)$
AUC	$\sum_{i=1}^n (FPR_{i+1} - FPR_i) \left(\frac{TPR_{i+1} + TPR_i}{2} \right)$
MCC	$\frac{(TP \times TN) - (FP \times FN)}{\sqrt{(TP + FP)(TN + FN)(TP + FN)(TN + FP)}}$
Brier loss	$\frac{1}{N} \sum_{i=1}^N (f_i - o_i)^2$
Precision	$\frac{TP}{TP + FP}$
F1	$2 \times \frac{\text{precision} \times \text{recall}}{\text{precision} + \text{recall}}$
Cohen kappa	$\kappa = \frac{p_o - p_e}{1 - p_e}$

TPR = True positive rate; FPR = False positive rate; N = No. of predictions; f_i = Predicted outcome; o_i = Actual outcome; p_o = Observed agreement; p_e = Expected agreement.

prediction of BBB classification [14,15]. Combination of *in vitro* and *in silico* models have also been explored using PAMPA data for developing predictive models with ML algorithms to examine the passive permeability of BBB for structurally related CNS (Central nervous systems) drug molecules [16]. Recently, Kato et al. [17] have reported on using

Random Forest, XGBoost, Histogram gradient boosting (HGB) and GCNN algorithms for developing QSAR models to predict brain penetration of drug candidates using an extensive PAMPA–BBB dataset with molecular descriptors and fingerprints. Apart from these aspects, *in silico* models using ML for skin permeability using PAMPA assay has also been discussed in various literatures [18–21]. It has been also used to prediction bio pharmaceutical classification system (BCS) at different levels of pH and screen chemicals of environmental concern through *in silico* approach [22,23].

The above discussion demonstrates the potential PAMPA's utility in assessment of different permeabilities – membrane, BBB, skin etc. It is an integral part of *in silico* studies related to drug development process due to its flexibility and adaptability. However, there are certain gaps – a generalized permeability prediction model using PAMPA, understanding of applicability domains, interpretable ML models etc. Also, majority of the works have used ML for predicting the permeability scores using PAMPA assays. The aim of the current study is to use the PAMPA permeability scores reported in different assays and develop general ML based models for predicting low permeability and moderate or high permeability of small molecules. Three classification ML models – Random forest (RF), Explainable boosting machine (EBM) and Adaboost algorithms have been trained with molecular descriptors in building the predictive models. Additionally, a Graph Attention Network (GAT) based model was also explored to analyze the predictive potential using molecular graphs apart descriptors. A set of 5447 small molecules or compounds was curated from different literature which reported PAMPA permeability scores for passive absorption, BBB and cell membrane permeability. This dataset was used to train and validate the ML and GAT models apart from which an external dataset was also used to screen and evaluate these model's performance on unseen data. All the ML models were interpreted using SHapley Additive exPlanation (SHAP) algorithm [24] for understanding the role of dominant features. Applicability domain of the models were also explored with Local outlier factor (LOF) algorithm [25] considering the structural diversity of the dataset. For further development and improvement of the models, the datasets and codes have been archived in GitHub (https://github.com/hridoy69/pampa_premeability) for reproducibility to abide by the FAIR



Scheme 1. A complete overview of the methodology used for developing machine learning and deep learning models for determining PAMPA permeable/non-permeable of compounds.

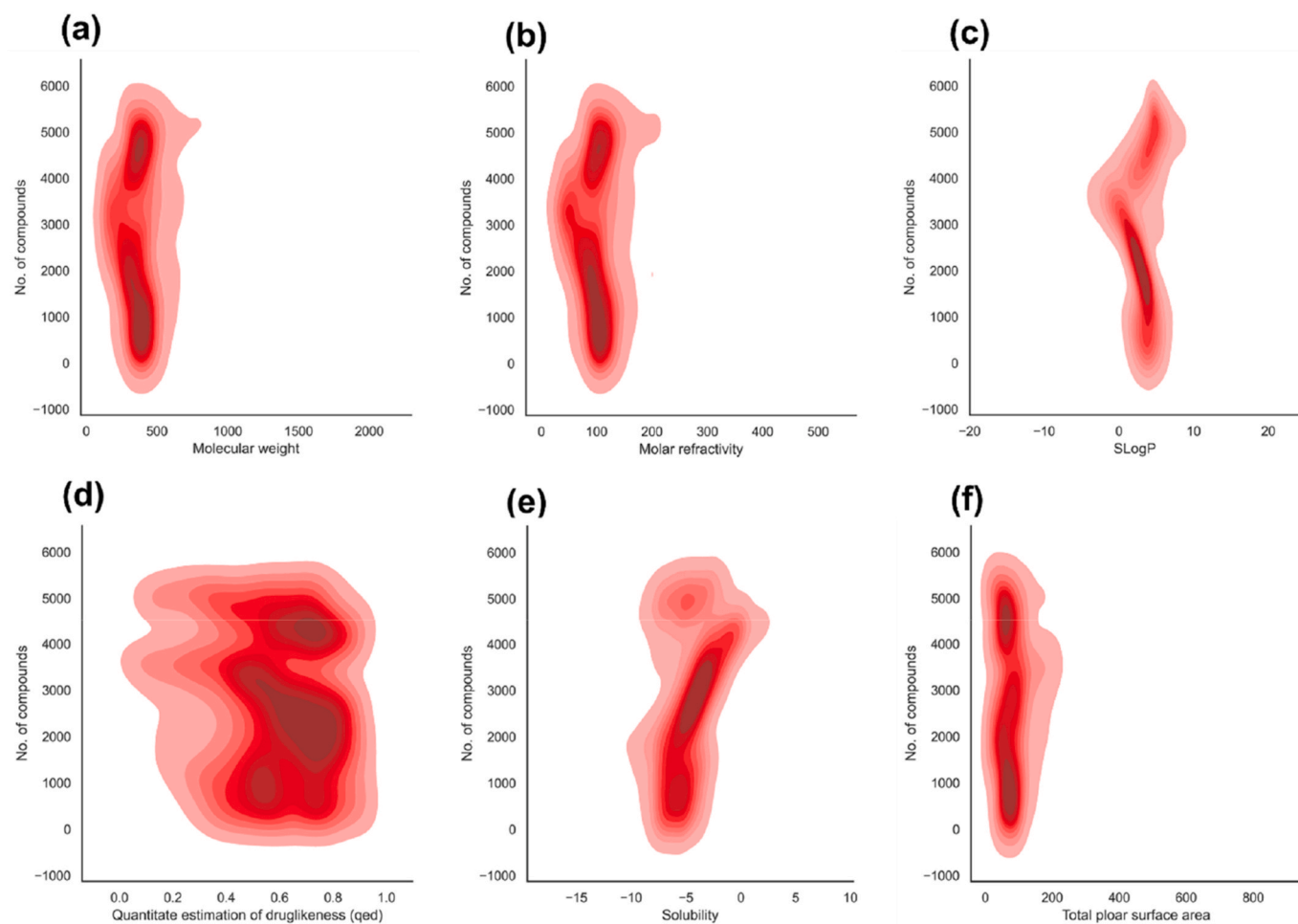


Fig. 1. Physicochemical property-based analysis of the dataset by comparing frequency versus property weights (a) molecular weight, (b) molar refractivity, (c) SlogP, (d) quantitative estimate of drug-likeness (qed) (e) solubility and (f) total polar surface area.

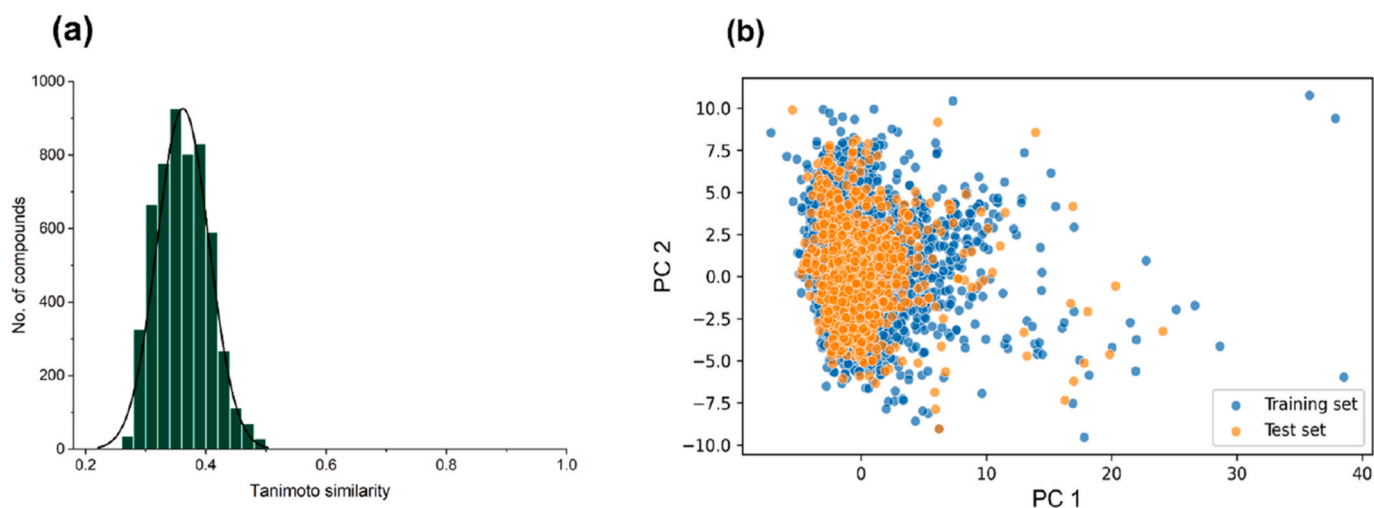


Fig. 2. (a) Structural diversity of the dataset calculated using Tanimoto similarity coefficient. It is seen that most of the compounds in the dataset set are structurally different from each other in the range of 0.26–0.5 offering stability of classification models. (b) Principal component analysis of the train and test set with cumulative variance between PC1 and PC2 of 28.75.

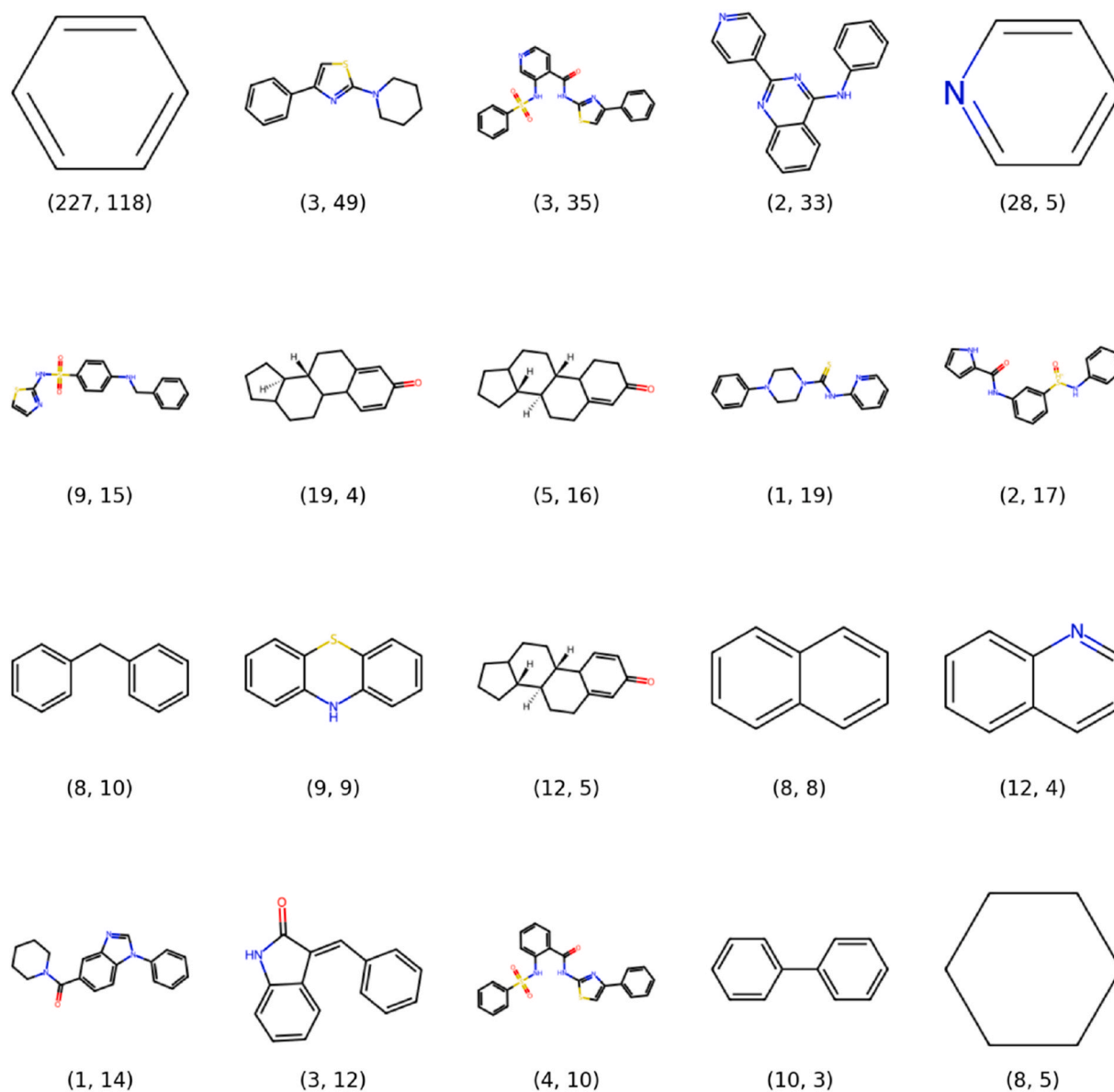


Fig. 3. Scaffold analysis of the dataset was performed using Bemis-Murcko analysis method. Structures of the most common scaffolds (top 20 based on frequency) for both the classes and their frequencies has been depicted below each structure. The first number resembles frequency of the scaffold among all the high and moderate PAMPA permeable (i.e. class 0) scaffolds in the dataset. Whereas, the second number resembles the frequency of a scaffold among all the low PAMPA permeable (i.e. class 1) compounds in the dataset.

Table 3

Performance analysis of all the models developed in this study in terms of different metrics for the hold-out test set.

Metrics	RF	EBM	Adaboost	GAT
Accuracy	0.81	0.80	0.76	0.74
Sensitivity	0.73	0.74	0.66	0.67
Specificity	0.88	0.86	0.84	0.80
Balanced accuracy	0.81	0.80	0.75	0.71
AUC	0.86	0.87	0.82	0.83
MCC	0.63	0.61	0.51	0.47
Brier loss	0.19	0.19	0.24	0.16
F1 score	0.78	0.78	0.71	0.71
Kappa	0.62	0.61	0.50	0.47

principles [26].

2. Materials and methods

2.1. Dataset generation

The dataset for this study has been curated from five different literature [4,8,12,17,27] with reported PAMPA permeability scores (in 10^{-6} cm/s) to determine passive absorptions, blood-brain barrier and cell membrane permeability. An initial set of ≈ 6300 compounds were collected from these works out of which redundant compounds and those with no permeability score were removed leaving a size of 5447 compounds. The effectiveness of a classifier is highly reliant on the definition of positive and negative samples. However, no universally accepted criterion for determining positive and negative samples exists in permeability prediction. Based on the previous studies [10], we have considered two cut-off values for assigning the class labels '0' and '1' such that compounds that have permeability value > 10 are designated

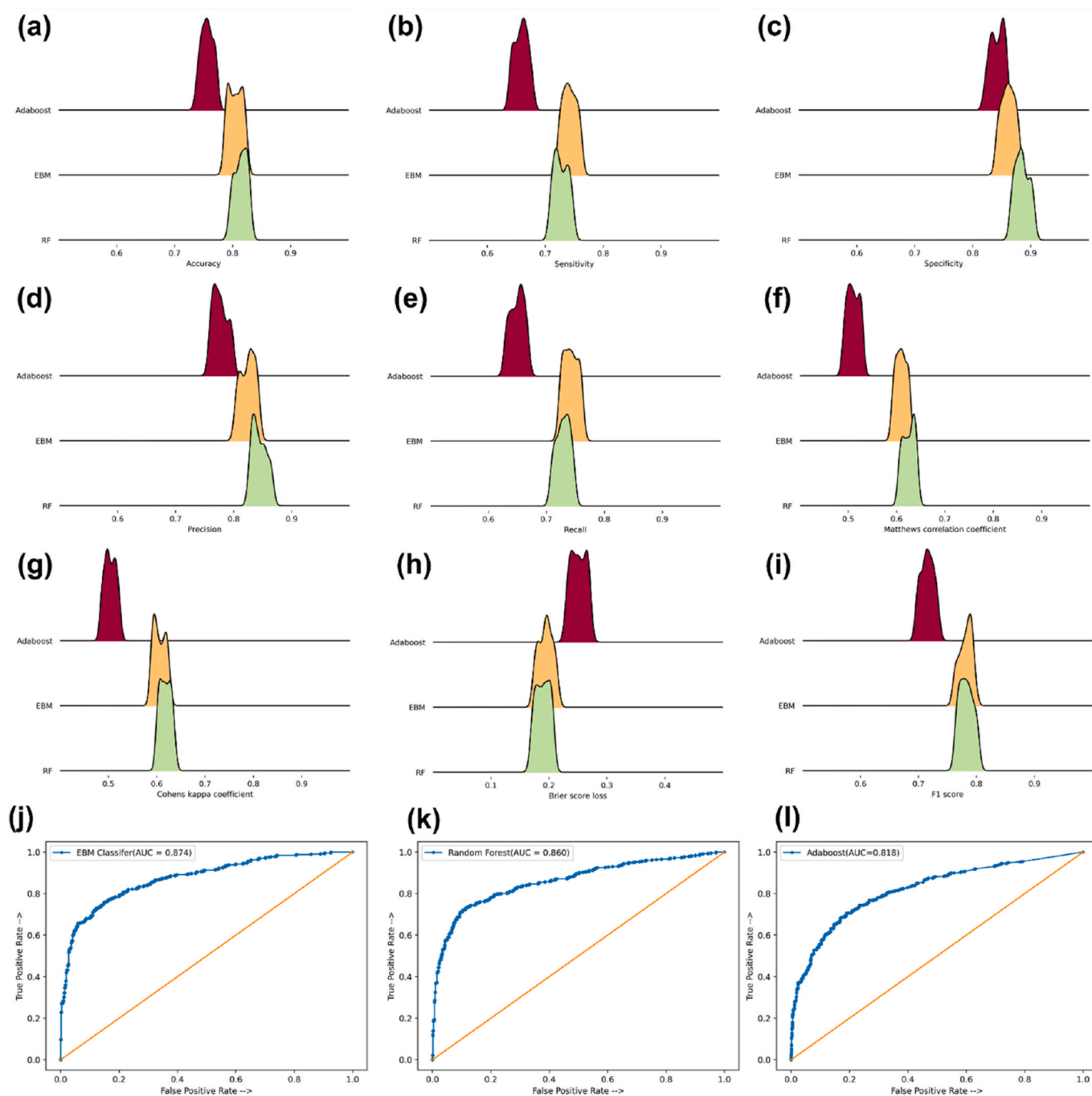


Fig. 4. Performance plots for the three models – EBM classifier, Random forest and Adaboost on different parameters (a) Accuracy, (b) Sensitivity, (c) Specificity, (d) Precision, (e) Recall, (f) MCC, (g) Cohen Kappa (h) Brier score loss, and (i) F1 score. ROC-AUC plots for the three models – (j) EBM classifier, (k) Random forest, and (l) Adaboost. EBM classifier predicted with the with highest AUC score of 0.87.

as ‘class 0’ which are ‘moderate to highly permeable’ and permeability value ≤ 10 categorized as 1 or ‘low permeable’ compounds. The current dataset has $\approx 53\%$ compounds belonging to class 0 and $\approx 47\%$ belonging to class 1, making it fairly balanced for training ML models. This final set was randomly split into 80–20 ratios for the purpose of training set and hold-out test set as shown in Table 1.

2.2. Scaffold analysis

To decipher the structural diversity of compounds in the dataset, molecular structures were examined through their scaffolds. The structures of the compounds belonging to both the classes (0 and 1) were used

to extract the Bemis-Murcko [28] scaffolds using the Scaffolds library of RDKit [29] in the Python environment. The scaffolds to molecule ratio was then calculated separately for both the permeable and non-permeable sets and also identifying the dominant scaffolds which gave atomistic level insights into their structures.

2.3. Molecular descriptors and feature selection

The chemical structures were used to generate molecular descriptors using RDKit [29] and Mordred [30] in the Python environment. A total of 1822 2D descriptors were generated by using both these tools which were then preprocessed by removing all attributes with large missing

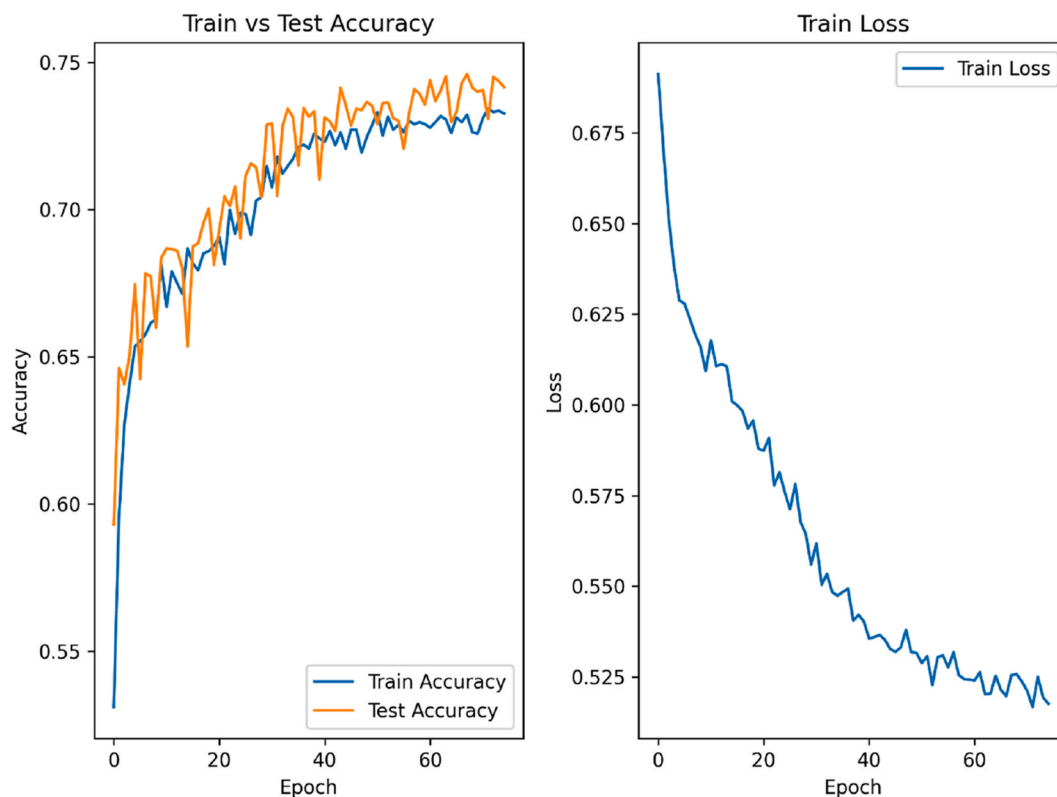


Fig. 5. Performance of the GAT model in terms of train accuracy and loss with test accuracy versus epochs. The number epoch was 75 which was derived through hyper parameter tuning using Optuna library.

Table 4

Comparative analysis of current study with previous reports on prediction of PAMPA permeability.

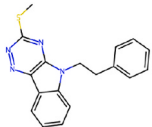
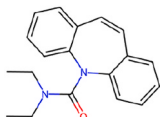
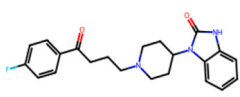
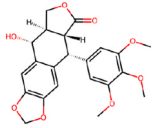
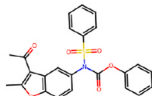
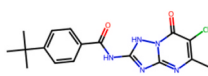
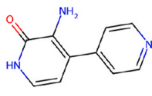
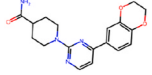
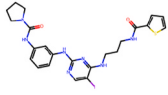
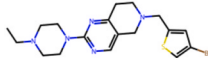
Works	Dataset size	Validated with external set?	Models used	AUC	BACC	Sensitivity	Specificity	Kappa
Siramshetty et al., 2021 [27]	16624	Yes (2664)	RF	0.90	0.82	0.78	0.86	0.64
			GCNN	0.90	0.82	0.81	0.84	0.65
Jordan Williams et al., 2022 [13]	6500	Yes (1046)	RF (RDKit)	0.83	0.74	0.62	0.87	0.41
			XGB (RDKit)	0.83	0.77	0.69	0.86	0.44
			DNN (RDKit)	0.80	0.70	0.50	0.90	0.37
			RF (MorganFP)	0.78	0.71	0.60	0.83	0.32
			XGB (MorganFP)	0.80	0.71	0.57	0.84	0.33
			DNN (MorganFP)	0.77	0.73	0.64	0.82	0.34
			GCNN	0.84	0.78	0.74	0.82	0.40
Kato et al., 2023 [18]	1958	No	RF (RdKit)	0.79	0.69	0.53	0.87	–
			XGB (RdKit)	0.78	0.68	0.46	0.90	–
			HGB(RdKit)	0.77	0.66	0.39	0.92	–
			GCNN (RDKit)	0.77	0.68	0.44	0.92	–
			RF (MorganFP)	0.74	0.66	0.45	0.86	–
			XGB (MorganFP)	0.73	0.63	0.36	0.90	–
			HGB(MorganFP)	0.72	0.61	0.30	0.92	–
			GCNN (MorganFP)	0.62	0.62	0.40	0.84	–
			RF (MOE)	0.79	0.68	0.45	0.91	–
			XGB (MOE)	0.79	0.69	0.49	0.89	–
			HGB(MOE)	0.78	0.69	0.48	0.90	–
			GCNN (MOE)	0.78	0.68	0.43	0.91	–
This study, 2024	5447	Yes (486)	RF	0.86	0.81	0.73	0.88	0.62
			EBM	0.87	0.80	0.74	0.86	0.61
			AdaBoost	0.82	0.75	0.66	0.84	0.50
			GAT	0.83	0.71	0.67	0.80	0.47

values, values as '0' or undefined and some which were commonly generated by both of them. Additionally, a significant portion of descriptors calculated by Mordred, particularly autocorrelations, tended to be highly correlated or redundant and were therefore excluded from being in the dataset. Finally, at this phase, a dataset of 640 molecular descriptors for these 5447 compounds was retained for feature selection. The first step towards feature selection and refining the dataset was to

the recursive feature elimination (RFE) method. RFE works by recursively eliminating the least important features and building a model on the remaining features until the desired accuracy is achieved. In this study, RFE was applied over 3 iterations to reach at 100 important features which were ranked based on their feature importance scores. To avoid highly correlated features in the dataset, it was further subjected to Spearman's rank correlation to arrive at a more robust dataset for

Table 5

Results for a set of compounds from Pubchem bioassay (AID 1645871) used as external dataset to validate with the best models in this study – random forest, explainable boosting machine classifier and GAT based model.

Pubchem ID	Labels	Probabilities (Predicted)	2D Structures
563972	Permeability: >1000 True: 0 Predicted: 0	RF: 0.83 EBM: 0.91 GAT: 0.83	
766793	Permeability: >1000 True: 0 Predicted: 0	RF: 0.79 EBM: 0.63 GAT: 0.79	
16363	Permeability: 98.19 True: 0 Predicted: 0	RF: 0.81 EBM: 0.92 GAT: 0.86	
10607	Permeability: 262 True: 0 Predicted: 0	RF: 0.77 EBM: 0.94 GAT: 0.72	
1725261	Permeability: >1000 True: 0 Predicted: 0	RF: 0.76 EBM: 0.89 GAT: 0.76	
6485805	Permeability: 83.87 True: 0 Predicted: 0	RF: 0.78 EBM: 0.67 GAT: 0.88	
25181364	Permeability: 2.72 True: 1 Predicted: 1	RF: 0.69 EBM: 0.71 GAT: 0.61	
46907663	Permeability: 190.7 True: 0 Predicted: 0	RF: 0.90 EBM: 0.92 GAT: 0.77	
10077147	Permeability: 233.72 True: 0 Predicted: 0	RF: 0.83 EBM: 0.82 GAT: 0.94	
16028187	Permeability: 77.11 True: 0 Predicted: 0	RF: 0.86 EBM: 0.82 GAT: 0.93	

training the models.

2.4. Predictive models and validation

Three ML algorithms that are conceptually different from each other have been used in this study for the classification of PAMPA permeability. a) First is RF, a widely used ensemble learning algorithm in classification due to its high accuracy and robustness [31]. It is built with multiple decision trees where each tree is trained with a random subset of the dataset making the model highly stable as well as avoid overfitting. b) Second is EBM which is a typical Generalized additive model (GAM) that offers both interpretability as well as high predictive performance. Although they are often accurate as state-of-the-art black box ML models, but remained completely interpretable [32]. c) The third model is Adaboost, which also follows ensemble learning but

combines multiple weak learners into a single strong learner to improve the predictive accuracy [33]. The term ‘adaptive’ in AdaBoost often reflects its ability to adapt to errors by altering the weights of misclassified instances, and effectively boosting the performance.

All these models were implemented in Python using Scikit-Learn [34, 35] and related libraries. The trained models were subjected to k-fold cross-validation to assess their performance and ensure that they can be generalized well to unseen data. Besides this, the overall performance of classification models was evaluated mainly on the True Positive (TP), True Negative (TN), False Positive (FP), and False Negative (FN) results and a set of corresponding metrics as depicted in Table 2.

2.5. Exploring GAT model

Deep learning (DL) has made commendable progress in different areas of drug discovery. Graph convolutional neural networks (GCNN) consider the chemical structures of compounds as input which are represented in a molecular graph such that the nodes signify atoms and edges signify bonds between the atoms [36]. GCNN has been prominent in building models for predicting ADMET properties, pharmacological activities, drug-target interactions, and more [4,15,27,37–40]. Graph attention networks [41] are extensions of GCNN which uses an attention mechanism to adaptively assign an importance (or ‘attention’) to different neighbouring nodes in the network. Each node in GAT are treated equally, and the attention scores or coefficient (e_{ij}) between the node and each of its neighbours are computed by,

$$e_{ij} = a \left(W \vec{h}_i, W \vec{h}_j \right) \quad (1)$$

$$\alpha_{ij} = \text{softmax}_j(e_{ij}) = \frac{\exp(e_{ij})}{\sum_{k \in N_i} \exp(e_{ik})} \quad (2)$$

Where, W is the weight matrix, N_i is a random neighbourhood, and softmax function is used to normalize the values for ease of comparison across the network. In this study, we have built a GAT based model with two convolution layers and two fully connected layers for classification. ReLU and Sigmoid functions were used for non-linearity and generate the output and probability scores respectively. Each node in the molecular graph represented an atom in the molecule that are described by four node features – atomic number, degree, formal charge and aromaticity. The optimal hyperparameter were selected with Optuna, a python library (Table S2). Similar to the ML models, the performance of GAT was also validated using the parameters as in Table 1. A detailed overview of the methodology followed in this study has been depicted in Scheme 1.

3. Results and discussion

3.1. The dataset

The dataset used in this study comprises 5447 compounds, with 4357 used as a training set and 1090 as a hold-out test set. The splitting was randomly done using the ‘train-test-split’ function in Scikit-learn package. To assess the spread of the dataset, we examined several physico-chemical properties, such as molecular weight (MW), molar refractivity (MR), sLogP, quantitative estimate of drug-likeness (qed), solubility, and topological polar surface area (TPSA). The compounds exhibited a broad range across these properties: 76.05–2181.32 g/mol for MW with standard deviation (SD) of 18.33, 1.37–539.80 m³/mol for MR with SD of 3.14, 0–901.57 Å² for TPSA with SD of 12.20, –17.81 to 8.56 for solubility with SD 2.35, –18.35 to 22.61 for sLogP with SD 1.13, and 0.01–0.94 for qed with SD 0.02 (Table S1) which have been graphically shown in Fig. 1. We have also assessed the structural diversity of the dataset using Tanimoto Similarity (TS) which shows the similarity score

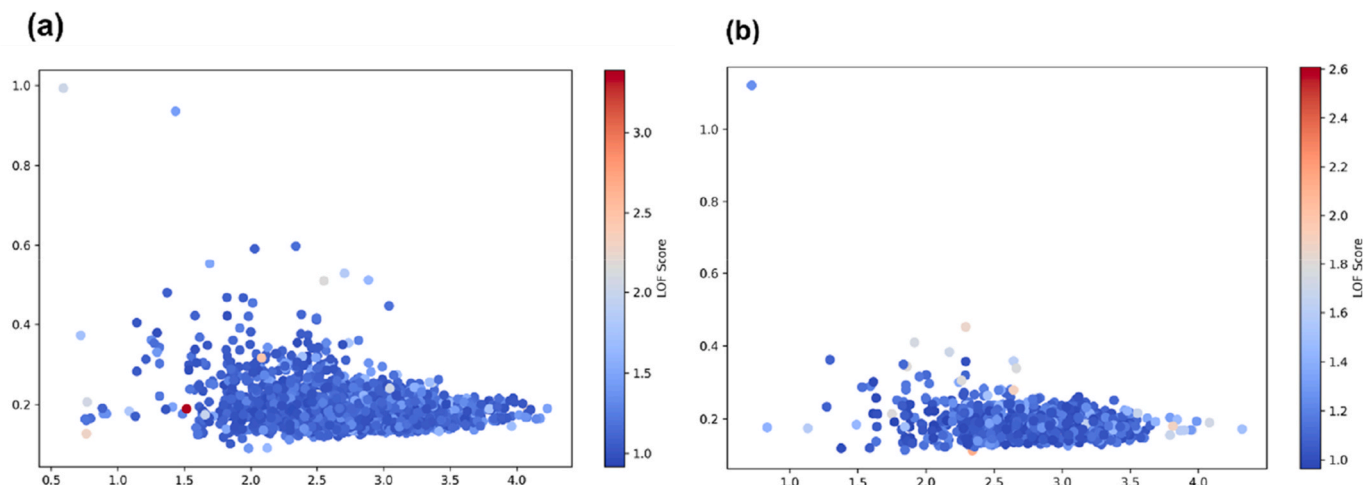


Fig. 6. Model AD in training sets (a) and test sets (b) with no of neighbourhood (k) = 5. The density of each point p is compared with its five neighbouring points to identify the outliers. The lower density points are mainly outside the AD whereas others are within AD.

lies between 0.26 and 0.5 and can be seen in Fig. 2 (a). Additionally, principal component analysis (PCA) was performed to confirm that the dataset splitting introduced no bias and Fig. 2 (b) demonstrates that the cumulative explained variance percentages of the first two principal components (PC1 and PC2) was 28.75 %, which indicates that both sets are evenly distributed in the chemical space.

3.2. Bemis – Murcko scaffold

To analyze and compare the core structures of the compounds belonging to both the classes (i.e. class 0 & class 1) we employed the Bemis – Murcko analysis. The scaffolds in these sets displayed significant diversity, from simple single-ring aromatics to more complex structures with 3 to 4-ring systems. The molecule–scaffolds ratio for compounds belonging to class 0 was found to be 0.76 whereas for class 1 was 0.47. In Fig. 3, we have shown the 20 most common or dominant scaffolds found in the compounds of both these classes as identified through this analysis.

3.3. Feature selection

RDKit and Mordred could generate a total of 1822 2D descriptors out of which after initial preprocessing and removal of autocorrelation descriptors, there were 640 descriptors or attributes which were subjected to RFE. After 3 iterations, RFE retained 100 descriptors as the most relevant and important features for model building. Thereafter, Spearman's rank correlation was computed for these 100 features to further eliminate the most highly correlated features (with a threshold of 0.8) which finally resulted in 56 features that were considered as final features for model building. The correlation matrix generated from Spearman's rank method for the initial 100 descriptors and the final 56 descriptors or features is shown in Fig. S2.

3.4. Model training and validation

Three ML — EBM, RF and AdaBoost, and one DL model GAT were trained to predict the PAMPA permeability of the compounds. The ML models were developed in Python using the Scikit-learn package. The GAT based model was built using Python libraries like PyTorch, PyTorch Geometric, RDKit and various Scikit-learn packages were used. To address overfitting and optimize hyperparameters for better accuracy, the models were initially validated through 10-fold cross-validation (Fig. S2). The validation results on the hold-out test set showed that the accuracy of the three models ranged between 0.75 and 0.81, with the

highest score of 0.81 for RF, and the lowest score of 0.75 by AdaBoost. In terms of AUC score, RF had the highest with 0.86 followed by EBM with 0.80 and Adaboost at 0.81. In comparison to ML models, the performance of GAT model was somewhat lower but near to the Adaboost model. The accuracy obtained with this model was around 0.74 with an AUC of 0.83. Many other parameters were evaluated such as sensitivity, specificity, balanced accuracy, MCC, precision, recall etc. whose detailed results have been shown in Table 3 and graphically in Figs. 4 and 5. Based on all these findings we can conclude that all three ML models and the GAT model performed significantly well in predicting the PAMPA permeability. Further, we compared our results with few of the recent works (Table 4) which used descriptors, fingerprints and molecular graphs in their models and it was seen that the current models performed better than the models developed using different fingerprints by Williams et al. [22] and Kato et al. [18] and was marginally less than that of Siramshetty et al. [26]. All these previous models used GCNN to train the molecular graphs however, the current study uses GAT.

3.5. Screening external dataset

An external dataset consisting of 486 oral drugs was obtained from Ref. [13] which was reported in Pubchem bioassay (AID_1645871) and used to validate all the models — RF, EBM, AdaBoost, and GAT developed in this study. These compounds had their respective PAMPA permeability score in 10^{-6} cm/s with phenotypes as low, high and moderately permeable. While testing the models with these compounds, it was found that the RF model could classify the PAMPA permeability/non-permeability with an accuracy of 91 %, EBM with 90 %, AdaBoost with 89 % and finally GAT with 86 %. When compared to performance previous works (as in Table 4) on external dataset, it was found that the accuracy of the models in Ref. [22] was in the range of 77 %–84 % and that in Ref. [26] was 82 % which shows that the models reported in this study not only performed well with the validation set but better with the external dataset. In Table 5, we have shown 10 random compounds from the external set with their observed PAMPA permeability score, actual class and predicted class. These compounds were predicted correctly by RF, EBM and GAT based model with their respective prediction probabilities. Further, all these compounds were also subjected to Bemis – Murcko scaffolds analysis which reported 149 unique scaffolds. When compared with the scaffolds of the dataset used in this study it was found that, 79 of these scaffolds matched with those for class 1 and 81 for class 0. Twenty random scaffolds for both these classes from the external dataset has been shown in Fig. S6. These findings validate that the model's reported in this study is able to

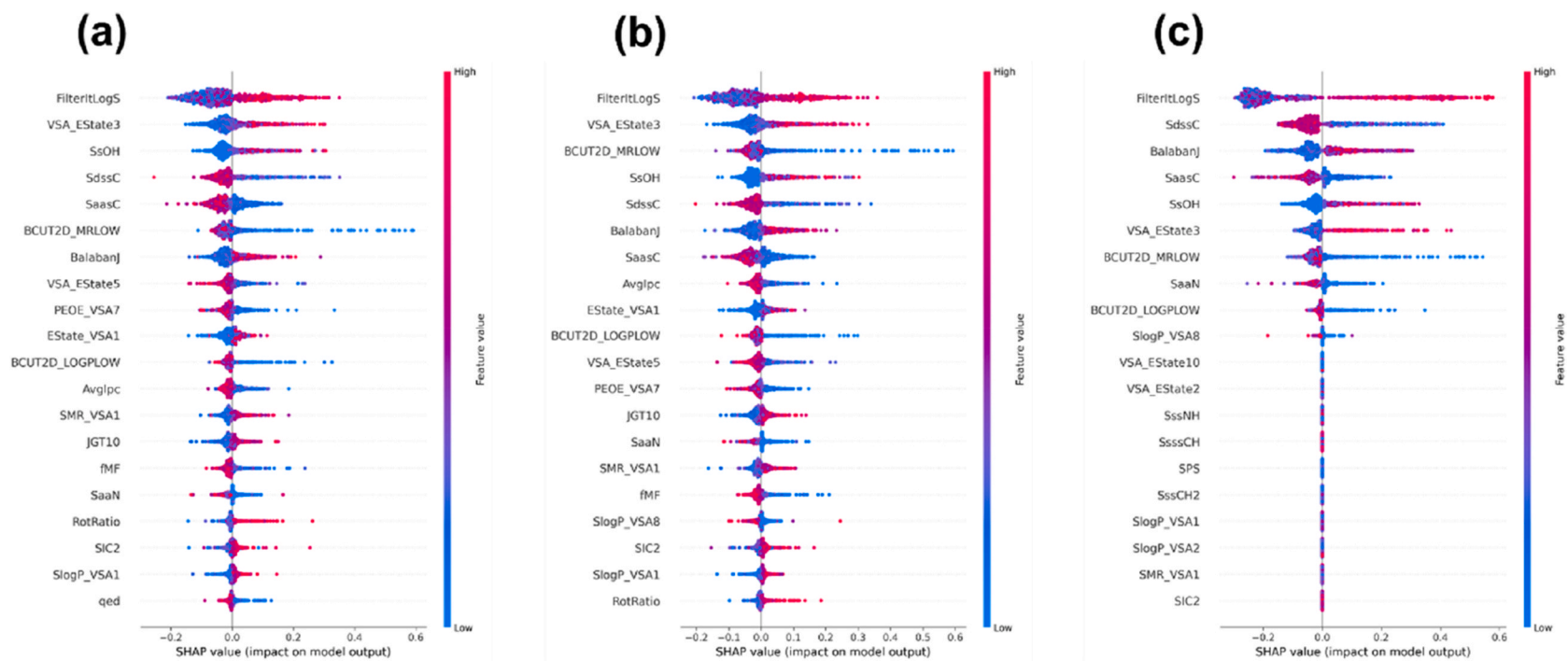


Fig. 7. Summary of the SHAP plot with important features for explainability of the models – (a) EBM classifier, (b) Random Forest, and (c) Adaboost.

capture some key structural features that influence permeability.

3.6. Applicability domain and model interpretation

From ML perspectives, applicability domain (AD) plays an important role to prevent prediction deviation by a large gap specially when the structural diversity of the training set is limited. We used the Local outlier factor (LOF) algorithm [24] to detect the applicability domain compounds from the training set. The LOF module in scikit-learn was applied to obtain the LOF scores for each compound which help to find the compounds which were within the domain and outside the domain. As seen in Fig. 6, there are few compounds which were outliers in the training set which depicts that large number of compounds were within the domain. Thus, selecting compounds which are similar to the ones in the training dataset may results into higher prediction accuracy when using the present model.

Explainable AI helps in understanding the learning process of ML models by analyzing the marginal impact of features during training. The SHapley Additive exPlanation (SHAP) algorithm based on game theory, used to compute feature importance scores, is now recognized and widely used as a unified approach for interpreting model predictions [25]. The SHAP dependence plot is often employed to assess the effect of individual features on prediction models. Among the 56 final features used in this current study for building the predictive model, some of the most influential were FilterItLogS, VSA_Estate3, BCUT2D_MRLOW, SsOH, Balabanj, JGT10, SlogP_VSA1, and qed which contributed in all the three ML models – RF, EBM and Adaboost. A detailed explanation of the most impactful features of each model is presented in Fig. 7.

4. Conclusion

PAMPA can be considered as a surrogate for preliminary assessment of drug absorption and permeability. In this study, we have used a dataset of 5447 compounds or small molecules which have been curated from different literatures. This dataset has been used to train and validated three ML models – RF, EBM and Adaboost to classify PAMPA permeable and non-permeable compounds using molecular descriptors. Apart from these models, GAT was also implemented for building predictive models. All these models were assessed with a set of metrics which revealed that, among the three ML models, RF and EBM had a similar predictive score of 81 % and 80 % respectively and Adaboost with 76 %. The GAT-based model however achieved an accuracy of 74 %. All these models have also subjected to screening an external dataset of 486 compounds and the predictive accuracy of the three models lay between 89 %–91 %. Interestingly, when the GAT model was used to screen the same external set, the predictive accuracy was around 86 %. These results demonstrate that the models developed in this study to predict PAMPA permeability performs significantly well on the benchmarks covering diverse chemical space and can promote medicinal chemists in using them to efficiently prioritize compounds in drug development.

CRediT authorship contribution statement

Upashya Parasar: Writing – original draft, Validation, Formal analysis, Data curation. **Orchid Baruah:** Writing – original draft, Validation, Formal analysis, Data curation. **Debasish Saikia:** Writing – review & editing, Writing – original draft, Visualization, Validation, Investigation, Formal analysis. **Pankaj Bharali:** Writing – review & editing, Writing – original draft, Visualization, Validation. **Hridoy Jyoti Mahanta:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Methodology, Investigation, Conceptualization.

Data availability

All the data and the codes associated with this work is available for reproduce in GitHub. https://github.com/hriday69/pampa_premeability

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used no generative AI and AI-assisted technologies.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Hridoy Jyoti Mahanta reports administrative support and infrastructure were provided by CSIR - North East Institute of Science and Technology. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Abbreviations

AUC, Area under the curve; BBB, Blood–brain barrier; CNS, Central nervous systems; DL, Deep learning; EBM, Explainable boosting machine; FAIR, Findable, accessible, interoperable and reusable; FP, False positive; FN, False negative; GCNN, Graph convolution neural network; GAM, Generalized additive model; GAT, Graph attention network; ML, Machine learning; MAE, Mean absolute error; MW, Molecular weight; MR, Molar refractivity; PAMPA, Parallel artificial membrane permeability assay; PCA, Principal component analysis; qed, quantitative estimate of drug-likeness; QSAR, Quantitative structure–activity relationship; RF, Random forest; RFE, Recursive feature elimination; SGD, Stochastic gradient decent; SMILES, Simplified molecular input line entry system; SHAP, Shapley additive explanation; SD, Standard deviation; TS, Tanimoto similarity; TPSA, Topological polar surface area; TP, True positive; TN, True negative.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jmngm.2025.109050>.

Data availability

Data is available on github

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